

Reviewer Pack — Minimal Local Ring Unit Group Size and Communication Complex...

Entry f9128187d402 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Ring Unit Group Size and Communication Complexity Rank Correlation via Local Class Group

Entry ID: f9128187d402 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 04:05:59 UTC

1. Conjecture

Field A (mathematical branch): Local Ring Theory **Field B** (complexity object): Communication Complexity

Statement:

The local class group size of a finite field extension K/L , where L is the base field for a communication protocol, is correlated with the rank of the communication complexity for that protocol, such that $|Cl(K/L)| = \Theta(ccr(K/L))$.

Rationale (proposer's reasoning):

Local ring theory provides tools to analyze algebraic structures that can capture the intricacies of information flow in distributed computations. The local class group measures the failure of a ring to be a product of a finite number of maximal ideals and could potentially provide insights into the complexity of communication protocols.

Taxonomy category: local_ring_theory_communication_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 21fa2aeb8108f900

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Correlation between local class group size $|Cl(K/L)|$ and communication complexity rank $ccr(K/L)$ of finite field extensions K/L is within ± 0.05 standard deviation from a Pearson correlation coefficient of 1 using 30 random seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.90	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - size local class group finite field extension AND communication complexity rank - correlation local class group size finite field extension AND communication complexity rank - $|Cl(K/L)|$ proportional to $ccr(K/L)$ in local ring theory AND communication complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def is_prime(n):
        if n <= 1:
            return False
        for i in range(2, int(math.sqrt(n)) + 1):
            if n % i == 0:
                return False
        return True

    def generate_primitive_element(p):
        while True:
            a = random.randint(2, p - 1)
            if all(pow(a, (p - 1) // q, p) != 1 for q in range(2, int(math.sqrt(p)) + 1)):
                return a

    def local_class_group_size(p):
        if not is_prime(p):
            raise ValueError("p must be a prime number")
        return p - 1

    def communication_complexity_rank(n):
        # Placeholder function for the actual computation
        # This should be replaced with the actual algorithm for computing ccr(K/L)
        return random.randint(1, n)

    n_values = [5, 10, 15, 20, 30, 40]
    instances_tested = 0
```

```

total_correlation = 0.0

for n in n_values:
    for _ in range(5):
        p = random.randint(n + 1, 2 * n)
        if not is_prime(p):
            continue
        alpha = generate_primitive_element(p)
        K = [alpha ** i % p for i in range(p)]
        L = [i % p for i in range(p)]

        try:
            cl_size = local_class_group_size(p)
            ccr = communication_complexity_rank(n)
            instances_tested += 1
            total_correlation += abs(cl_size - ccr) / (cl_size + ccr)
        except Exception as e:
            return {
                "metric_name": "Correlation between local class group size and communication complexity rank",
                "metric_value": None,
                "instances_tested": instances_tested,
                "n_max": n,
                "conjecture_holds": False,
                "counterexample": str(e)
            }

if instances_tested < 30:
    return {
        "metric_name": "Correlation between local class group size and communication complexity rank",
        "metric_value": None,
        "instances_tested": instances_tested,
        "n_max": max(n_values),
        "conjecture_holds": False,
        "counterexample": "Insufficient instances tested"
    }

mean_correlation = total_correlation / instances_tested
return {
    "metric_name": "Correlation between local class group size and communication complexity rank",
    "metric_value": mean_correlation,
    "instances_tested": instances_tested,
    "n_max": max(n_values),
    "conjecture_holds": abs(mean_correlation - 1) <= 0.05,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
    results = [run_trial(seed) for seed in seeds]

    mean_metric_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])

```

```
print(f"RESULT: {'SUPPORTED' if support_fraction >= 0.8 else 'FALSIFIED'} mean={mean_metric_va
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_20099062.py", line 102, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_20099062.py", line 102, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
KeyError: 'seed'
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which prevents us from verifying the correlation between the local class group size and communication complexity rank. | next: Re-run the test to ensure it completes successfully and provides the necessary data for analysis.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13006
2	propose	ollama_remote	glm4:latest	0	0	12212
3	preregistration	ollama_remote	glm4:latest	0	0	9163
4	novelty	ollama_remote	glm4:latest	0	0	8363
5	novelty	ollama_remote	glm4:latest	0	0	8931
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19731
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16110
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16860
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	24503
10	judge	ollama_remote	glm4:latest	0	0	9018

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 137897 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/f9128187d402.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/f9128187d402.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/f9128187d402.tar.gz (if generated)